Nazorat testi · *Aralash test · 16-variant* — *Milliy sertifikat 2026. Kimyo*

Vaqt: *180 daqiqa*

Davriy jadvaldan foydalanish mumkin.

1. Cl_2 gazining -23 °C va 2 atm bosimdagi **zichligini (g/l) aniqlang.** (R = 0,082 l·atm/(mol·K))
A) *13,85* B) *6,93* C) *3,46* D) *3,17*

2. 16,4 l CO_2 -23 °C va 0,5 atm bosimda necha gramm keladi? (R = 0,082 l·atm/(mol·K))
A) *35,2* B) *8,8* C) *17,6* D) *32,21*

3. Quyidagi moddalarning qaysilarida donor–akseptor bog‘ mavjud? 1) CO_2; 2) KNO_3; 3) NaOH; 4) $(NH_4)_2SO_4$
A) *3, 4* B) *2, 4* C) *2, 3* D) *4*

4. Reaksiya tezligining haroratga bog‘liqligi jadvalda berilgan:

T, °C	**20**	**30**	**40**
v, mol/(l·s)	0,5	1	2

Jadvaldan foydalanib, shu reaksiyaning 60 °C dagi tezligini (mol/(l·s)) **aniqlang.**
A) *8* B) *4* C) *16* D) *12*

5. $KClO_3$ tarkibida Cl ning oksidlanish **darajasini aniqlang.**
A) *+3* B) *+2* C) *+5* D) *+4*

6. 8 g NaOH va 5,48 g HCl aralashtirilib, reaksiya o‘tkazildi ($NaOH + HCl \rightarrow NaCl + H_2O$). Hosil bo‘lgan NaCl **massasini (g) aniqlang.**
A) *11,7* B) *17,55* C) *4,39* D) *8,78*

7. Reaksiya tezligining haroratga bog‘liqligi jadvalda berilgan:

T, °C	**20**	**30**	**40**
v, mol/(l·s)	0,5	1	2

Jadvaldan foydalanib, shu reaksiyaning 50 °C dagi tezligini (mol/(l·s)) **aniqlang.**
A) *2* B) *8* C) *4* D) *3*

8. 1 M H_2SO_4 eritmasining normal **konsentratsiyasini (n) aniqlang.**
A) *1* B) *2* C) *4* D) *0,5*

9. Jadvalda Cl elementining birikmalaridagi oksidlanish darajasi berilgan:

Modda	**Oksidlanish darajasi**
$HClO_4$	+7
HClO	+1
NaCl	-1
$KClO_3$	+5

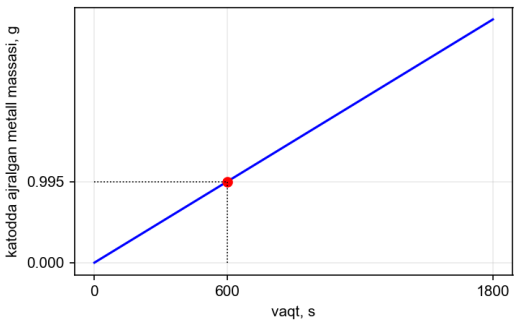
Ushbu moddalardan nechtasida Cl elementi faqat oksidlovchi xossani namoyon qiladi? (Cl uchun eng past oksidlanish darajasi -1, eng yuqorisi +7)
A) *2* B) *3* C) *0* D) *1*

10. $H_2 + I_2 \rightleftarrows 2HI$ reaksiyada muvozanat qaror topganda $[H_2] = 1$, $[I_2] = 2$, $[HI] = 2$ mol/l bo‘ldi. I_2 ning boshlang‘ich **konsentratsiyasini (mol/l) aniqlang.**
A) *2* B) *3* C) *4* D) *6*

11. Quyidagi tuzlarning qaysilari suvli eritmasi neytral muhitga ega? 1) $ZnCl_2$; 2) $NaNO_2$; 3) K_2SO_4; 4) NH_4Cl
A) *3* B) *1, 3* C) *1, 4* D) *2*

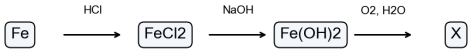
12. 13,9 g $FeSO_4 \cdot 7H_2O$ 200 g suvda eritildi. Eritmadagi tuzning massa **ulushini (%) aniqlang.**
A) *3,8* B) *6,5* C) *3,55* D) *7,11*

13. Doimiy tok kuchida Cu tuzining eritmasi elektroliz qilinganda katodda ajralgan metall massasining vaqtga bog‘liqligi grafikda berilgan (M(Cu) = 64 g/mol, valentlik 2).



Xuddi shu sharoitda 1800 s da ajralgan metall **massasini (g) toping.**
A) *5,969* B) *1,492* C) *2,984* D) *1,99*

14. Rasmdagi o‘zgarishlar zanjirida *X* moddaning **formulasini aniqlang.**



A) *$Fe(OH)_3$* B) *$FeCl_2$* C) *SO_3* D) *$Fe(OH)_2$*

15. Cu → ? → ? → ? o‘zgarishlar zanjirida (reagentlar: O_2, H_2SO_4, NaOH) 1-bosqichda hosil bo‘ladigan modda **formulasini toping.**
A) *Na_2SO_3* B) *$Ca(OH)_2$* C) *NaCl* D) *CuO*

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16. $NH_3 + H_3PO_4 \rightarrow NH_4H_2PO_4$ jarayoni uchun 4,48 l (n.sh.) NH_3 olingan (unum 90%). $NH_4H_2PO_4$ **miqdorini (mol) aniqlang.**
A) *0,2* B) *0,36* C) *0,09* D) *0,18*

17. $Cu(NO_3)_2$ qizdirilganda qanday moddalar hosil bo‘ladi?
A) *Ag, NO_2 va O_2* B) *ZnO, NO_2 va O_2*
C) *CuO, NO_2 va O_2* D) *N_2O va H_2O*

18. $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$ reaksiyasida 12 g FeS_2 ishtirok etdi. SO_2 **hajmini (l, n.sh.) toping.**
A) *1,12* B) *4,48* C) *8,96* D) *2,24*

19. 17 g $AgNO_3$ to‘liq parchalanguncha qizdirildi. Qattiq qoldiq **massasini (g) aniqlang.**
A) *10,8* B) *17* C) *21,6* D) *5,4*

20. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ jarayoni uchun 8,96 l (n.sh.) C_3H_8 olingan. CO_2 **hajmini (l, n.sh.) aniqlang.**
A) *26,88* B) *2,99* C) *53,76* D) *13,44*

21. CH_3–$CH(CH_3)$–$CH(C_2H_5)$–CH_2–CH_3 birikmaning sistematik nomi qaysi qatorda to‘g‘ri keltirilgan?
A) *2-metil-3-etilpentan* B) *3-etil-2-metilpentan*
C) *3-izopropilpentan* D) *2,3-dietilbutan*

22. 9,2 g toluoldan benzoy kislota olishda unum 75% bo‘lsa, mahsulot **massasini (g) aniqlang** (boshlang‘ich modda massasi 9,2 g).
A) *12,2* B) *18,3* C) *9,15* D) *6,9*

23. $C_2H_4 + H_2O$ (H^+) reaksiyasi natijasida qaysi organik modda hosil bo‘ladi?
A) *metan* B) *etanol*
C) *etilatsetat* D) *benzoy kislota*

24. 44 g CH_3CHO dan quyidagi ikki bosqichda C_2H_4 olinadi: 1) $CH_3CHO + H_2 \rightarrow C_2H_5OH$ (unum 75%); 2) $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (unum 75%). Oxirgi mahsulot C_2H_4 ning **hajmini (l, n.sh.) toping.**
A) *12,6* B) *16,8* C) *22,4* D) *25,2*

25. $C_4H_{10}O$ oddiy efirlarning izomerlari soni nechta?
A) *4* B) *3* C) *2* D) *5*

26. 17,6 g $CH_3COOC_2H_5$ dan quyidagi ikki bosqichda Na_2CO_3 olinadi: 1) $CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$ (unum 75%); 2) $CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$ (unum 90%). Oxirgi mahsulot Na_2CO_3 ning **massasini (g) toping.**
A) *19,08* B) *15,9* C) *21,2* D) *14,31*

27. Uglevodlar haqida qaysi fikrlar to‘g‘ri? 1) Glyukoza "kumush ko‘zgu" reaksiyasini beradi; 2) Selluloza β-glyukoza qoldiqlaridan iborat; 3) Fruktoza ketospirt; 4) Glyukoza gidrolizlanadi
A) *1, 2, 3* B) *2, 3, 4* C) *3, 4* D) *2, 3*

28. Aminlar, aminokislotalar va oqsillar haqida qaysi fikrlar to‘g‘ri? 1) Oqsillar biuret reaksiyasini beradi; 2) Nitrobenzoldan anilin Zinin reaksiyasi bilan olinadi; 3) Dipeptidda 2 ta peptid bog‘ bor; 4) Aminlar kuchsiz asoslar xossasini namoyon qiladi
A) *3, 4* B) *2, 3* C) *1, 3, 4* D) *1, 2, 4*

29. Yuqori molekulyar birikmalar haqida qaysi fikrlar to‘g‘ri? 1) Kapron poliamid tola; 2) Polimerning M = zveno M · polimerlanish darajasi; 3) Teflon — polivinilxlorid; 4) Fenolformaldegid smola polikondensatsiya mahsuloti
A) *1, 2, 4* B) *1, 2, 3* C) *2, 4* D) *1, 4*

30. 100 g 30% li va 100 g 40% li NaOH eritmalari aralashtirildi. Yangi eritmaning massa **ulushini (%) toping.**
A) *70* B) *35* C) *31,5* D) *38,5*

31. Kimyo laboratoriyasida ishlash qoidalari haqida qaysi fikrlar to‘g‘ri? 1) Ishqor teriga tushsa, qo‘l suv bilan yuviladi; 2) Probirka qizdirilganda og‘zi odamga qaratilmaydi; 3) Zaharli gaz ajraladigan tajriba mo‘rili shkafda o‘tkaziladi; 4) Konsentrlangan kislotani suvga ohista, aralashtirib quyiladi
A) *2, 3, 4* B) *1, 3, 4*
C) *1, 2, 3, 4* D) *1, 2, 4*

32. Rasmdagi asbob bilan qaysi aralashmani ajratish mumkin?



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A) *suv va qum* B) *suv va etanol*
C) *osh tuzi eritmasi* D) *suv va kerosin* —

Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring. K_2CO_3 va $MgCO_3$ dan iborat 39 g aralashma mo‘l xlorid kislota bilan to‘liq reaksiyaga kirishdi.	A *292* B *8,96* C *35,95* D *28,76* E *17,92* F *584*
33. Ajralgan gaz hajmini (l, n.sh.) aniqlang.	
34. Aralashmani to‘liq eritish uchun sarflangan 10% li HCl eritmasining massasini (g) aniqlang (M(HCl) = 36,5 g/mol).	
35. Reaksiyadan so‘ng eritmadagi tuzlar umumiy massasini (g) aniqlang.	

36. $KMnO_4 + K_2SO_3 + H_2O \rightarrow MnO_2 + K_2SO_4 + KOH$ reaksiya tenglamasini tenglang va barcha koeffitsiyentlar **yig‘indisini aniqlang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

37. 8,5 g H_2O_2 reaksiyaga kirishdi ($2H_2O_2 \rightarrow 2H_2O + O_2$). Ajralgan O_2 **hajmini (l, n.sh.) hisoblang** (unum 80%).

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

38. 100 m^3 suvda $Ca(HCO_3)_2$ konsentratsiyasi 20 mol/m^3, $Mg(HCO_3)_2$ niki 20 mol/m^3. Suvni $Ca(OH)_2$ bilan to‘liq yumshatish uchun $Ca(OH)_2$ ning **massasini (kg) hisoblang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

39. 9,2 g etanol avval sirka aldegidgacha (unum 80%), so‘ng sirka kislotagacha (unum 60%) oksidlandi; hosil bo‘lgan kislota mo‘l etanol bilan efirlandi (unum 100%). Etilatsetat **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

40. 14,8 g propion kislota va 9,2 g etanol dan 60% unum bilan murakkab efir olindi. Efir **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

YOZMA ISH

41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

Tabiiy mis ikki izotopdan iborat: ^{63}Cu (70%) va ^{65}Cu (30%). 6,4 g shu element namunasi olingan ($Z = 29$, $N_A = 6{,}02 \cdot 10^{23}$).

1)Elementning o‘rtacha nisbiy atom massasini hisoblang.

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2)Namunadagi atomlarning umumiy miqdorini (mol) toping. Namunadagi protonlar sonini (mol) ham toping.
3)Og‘irroq izotop (^{65}Cu) atomlarining miqdorini (mol) hisoblang.
4)Namunadagi neytronlarning umumiy miqdorini (mol) toping. Bitta atom massasini (g) ham hisoblang.
5)Namunadagi atomlar sonini (10^{23} birligida) aniqlang. Namunadagi protonlar sonining neytronlar soniga nisbatini ham toping.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

42. Quyidagi o‘zgarishlar zanjirida X, Y, Z noma’lum organik moddalar:

$$C_2H_4Br_2 \xrightarrow{\text{KOH (spirtli)}} X \xrightarrow{\text{C (aktiv), 600 °C}} Y \xrightarrow{CH_3Cl,\ AlCl_3} C_6H_5CH_3$$

1)$C_2H_4Br_2 \rightarrow X$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
2)$X \rightarrow Y$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
3)$Y \rightarrow C_6H_5CH_3$: reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

6,2 g qizil fosfor mo‘l kislorodda yondirildi, hosil bo‘lgan oksid suvda to‘liq eritildi (ortofosfat kislota). Eritmaga 0,4 mol NaOH qo‘shildi.
1)Reaksiya tenglamalarini yozing. Hosil bo‘lgan P_2O_5 massasini (g) hisoblang.
2)Hosil bo‘lgan H_3PO_4 massasini (g) toping. Kislotada fosforning massa ulushini (%) ham hisoblang.
3)Eritmada qanday tuz hosil bo‘lganini aniqlang va uning massasini (g) hisoblang.
4)Hosil bo‘lgan eritmani to‘liq neytrallash uchun yana qancha NaOH (g) kerak ($Na_2HPO_4 \rightarrow Na_3PO_4$)?
5)Hosil bo‘lgan tuzning 10% li eritmasi massasini (g) hisoblang.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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Nazorat testi · *16-variant* **Javoblar kaliti**

Faqat o‘qituvchi uchun.

1–32-savollar

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
B	*C*	*D*	*A*	*C*	*D*	*C*	*B*	*D*	*B*	*A*	*C*	*C*	*A*	*D*	*D*
17	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**
C	*B*	*A*	*A*	*B*	*C*	*B*	*A*	*B*	*D*	*A*	*D*	*A*	*B*	*C*	*D*

33–35-savollar

No	Javob	Qiymat
33	*B*	8,96
34	*A*	292
35	*C*	35,95

36–40-savollar

No	To‘g‘ri javob
36	13
37	2,24
38	444
39	8,45
40	12,24

41–43-topshiriqlar: qisqa yechim

41-topshiriq.

1) $\mathrm{Ar} = (63{\cdot}70 + 65{\cdot}30)/100 = \mathbf{63{,}6}$

2) $\mathrm{n} = 6{,}4/63{,}6 = \mathbf{0{,}101\ mol}$; $\mathrm{n(p)} = Z{\cdot}\mathrm{n} = 29{\cdot}0{,}101 = \mathbf{2{,}918\ mol}$

3) $\mathrm{n}(^{65}\mathrm{Cu}) = 0{,}101{\cdot}30/100 = \mathbf{0{,}03\ mol}$

4) $\mathrm{N(neytron)} = \Sigma\,(\mathrm{A} - Z){\cdot}\text{ulush}$: $\mathbf{3{,}482\ mol}$; $\mathrm{m(atom)} = 63{,}6/(6{,}02{\cdot}10^{23}) = \mathbf{10{,}56{\cdot}10^{-23}\ g}$

5) $\mathrm{N} = \mathrm{n}{\cdot}\mathrm{N_A} = 0{,}101{\cdot}6{,}02{\cdot}10^{23} = \mathbf{0{,}61{\cdot}10^{23}}$; $\mathrm{n(p)/n(n)} = 2{,}918/3{,}482 = \mathbf{0{,}838}$

Jami: M 15 + A 10 = 25 ball.

42-topshiriq.

1) $\mathrm{C_2H_4Br_2} + \mathrm{KOH}$ (spirtli) $\rightarrow \mathrm{C_2H_2}$ (asetilen)

2) $\mathrm{C_2H_2} + \mathrm{C}$ (aktiv), 600 °C $\rightarrow \mathrm{C_6H_6}$ (benzol)

3) $\mathrm{C_6H_6} + \mathrm{CH_3Cl}$, $\mathrm{AlCl_3} \rightarrow \mathrm{C_6H_5CH_3}$ (toluol)

Jami 25 ball.

43-topshiriq.

1) $4\mathrm{P} + 5\mathrm{O_2} \rightarrow 2\mathrm{P_2O_5}$: $\mathrm{n(P_2O_5)} = 0{,}1$ mol $\rightarrow \mathbf{14{,}2\ g}$

2) $\mathrm{P_2O_5} + 3\mathrm{H_2O} \rightarrow 2\mathrm{H_3PO_4}$: n = 0,2 mol $\rightarrow \mathbf{19{,}6\ g}$; $\omega(\mathrm{P}) = 31/98 = \mathbf{31{,}6\%}$

3) $\mathrm{H_3PO_4} : \mathrm{NaOH} = 1 : 2 \rightarrow \mathbf{Na_2HPO_4}$ 0,2 mol = $\mathbf{28{,}4\ g}$

4) $\mathrm{Na_2HPO_4} + \mathrm{NaOH} \rightarrow \mathrm{Na_3PO_4} + \mathrm{H_2O}$: 0,2 mol · 40 = $\mathbf{8\ g}$

5) $\mathrm{m(eritma)} = 28{,}4/0{,}1 = \mathbf{284\ g}$

Jami: M 15 + A 10 = 25 ball.

Mavzular taqsimoti

No	Mavzu	Tekshiriladigan ko‘nikma	Ball
1	11-sinf (Mendeleyev–Klapeyron)	Hisoblash va masala yechish	1
2	11-sinf (Mendeleyev–Klapeyron)	Hisoblash va masala yechish	1
3	8-sinf; Tirkashev II bob (kimyoviy bog‘lanish)	Bilim va tushunish	1
4	8-sinf; Tirkashev VI bob	Jadval tahlili	1
5	8-sinf; Tirkashev *V* bob (oksidlanish darajasi)	Bilim va tushunish	1
6	8-sinf (ortiqcha modda masalasi)	Hisoblash va masala yechish	1
7	8-sinf; Tirkashev VI bob	Jadval tahlili	1
8	11-sinf 4-bob (normal konsentratsiya)	Hisoblash va masala yechish	1

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9	Tirkashev *V* bob (OQR, jadval)	Jadval tahlili	1
10	Tirkashev VI bob	Hisoblash va masala yechish	1
11	9-sinf (gidroliz); Tirkashev IV bob	Bilim va tushunish	1
12	Tirkashev III bob (kristallogidrat)	Hisoblash va masala yechish	1
13	11-sinf (Faradey qonuni, grafik)	Grafik tahlili	1
14	9-sinf (genetik bog‘lanish)	Rasm bo‘yicha tahlil	1
15	8-sinf; 9-sinf (genetik bog‘lanish)	Bilim va tushunish	1
16	9-sinf; Tirkashev (mineral o‘g‘itlar/nodir gazlar, hisob)	Hisoblash va masala yechish	1
17	Tashev (nitratlar)	Bilim va tushunish	1
18	9-sinf; Tirkashev ($S/SO_2/H_2SO_4$, hisob)	Hisoblash va masala yechish	1
19	Tashev (nitratlar termik parchalanishi)	Hisoblash va masala yechish	1
20	10-sinf (alkan, hisob)	Hisoblash va masala yechish	1
21	10-sinf (nomenklatura §6, §11, §20, §24–26)	Bilim va tushunish	1
22	10-sinf §16, §23, §33	Hisoblash va masala yechish	1
23	10-sinf (olinish usullari)	Bilim va tushunish	1
24	10-sinf (ikki bosqichli hisob)	Hisoblash va masala yechish	1
25	10-sinf §6, §12, §20, §28 (izomeriya)	Bilim va tushunish	1
26	10-sinf (ikki bosqichli hisob)	Hisoblash va masala yechish	1
27	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
28	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
29	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
30	Tirkashev III bob (eritmalar)	Hisoblash va masala yechish	1
31	7-sinf (xavfsizlik qoidalari); 10-sinf lab.	Bilim va tushunish	1
32	7-sinf (10-rasm)	Rasm bo‘yicha tahlil	1
33–35	Aralash masala (moslashtirish)	Ko‘p bosqichli hisob	3
36–40	Qisqa javobli masalalar	Hisoblash	5
41–43	Yozma ish	Usul va hisob (M, A)	75

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